Digital Twins in Medicine

Models, Complexity, and the Limits of the Virtual You

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A Brief History: From Apollo to the Patient



1960s — Apollo 13

Ground-based simulators with constant feedback saved three lives in space.



2000s — Industry

Michael Grieves formalized the DT concept for product lifecycle management.



Today — Healthcare

Patient-specific DTs for personalized medicine and precision health.

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Analogy for medical students

Think of Apollo 13: engineers on the ground had a *virtual replica* of the spacecraft updated with real-time telemetry to diagnose problems and test solutions *before* instructing the crew. That is the essence of a digital twin—and it is exactly what we want to do for a patient.

Models Are Everywhere in Medicine

❶ Anatomical Models
From Vesalius's drawings to 3D-printed organs—simplified representations of the body for teaching, diagnosis, and surgical planning.

❷ Statistical & Epidemiological Models
Framingham Risk Score, QRISK, SIR models for pandemics. They compress population data into actionable predictions.

❸ Mechanistic (Physiological) Models
Hodgkin–Huxley (nerve impulse), compartmental pharmacokinetics, cardiac electrophysiology. These encode *how* the body works.

❹ Machine Learning / AI Models
Pattern recognition at scale—retinal screening, pathology image classification, sepsis early warning systems.

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❺ Mental Models
Your clinical reasoning! Differential diagnosis is a mental model you continuously update with new data.

The Common Thread

Every model is a **simplification** of reality built for a **purpose**. No model captures everything. The question is always: *is this model fit for this purpose?*

What Does “Complexity” Actually Mean?

The human body is the ultimate complex system.

Clinical Example

Heart failure is not just a pump problem. It involves neurohormonal feedback (RAAS), renal compensation, inflammatory cascades, patient behavior, medication adherence, social support, psychological state—all interacting nonlinearly over time.

No DT can capture all of this.

Complicated vs. Complex

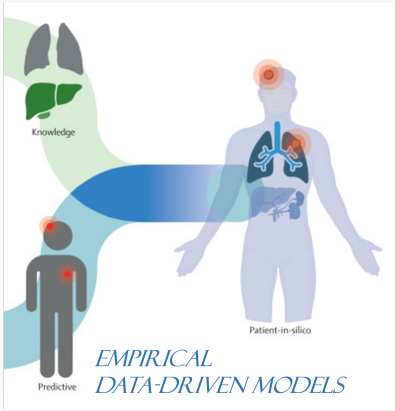
Complicated: A car engine—many parts, but predictable.
Complex: A patient—many parts, and *adaptive, emergent, unpredictable.*

Complex Systems are not only simply *complicated*, but:

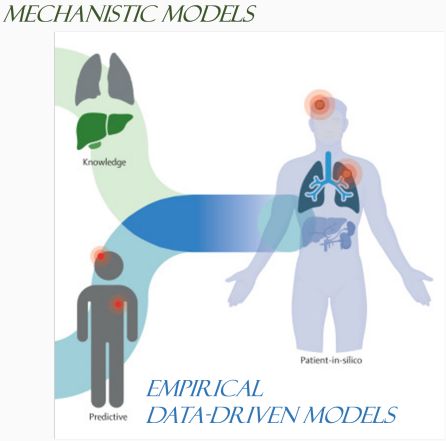
- **Many interacting parts** at multiple scales (molecular, cellular, organ, organism, social)
- **Nonlinear:** small changes can have large effects (and vice versa)
- **Emergent:** the whole has properties the parts do not
- **Feedback loops:** causes become effects and vice versa
- **Path-dependent:** history matters
- **Adaptive:** the system *changes in response to* your intervention

Mechanistic OR Empirical Models for Personalized Healthcare?

MECHANISTIC MODELS

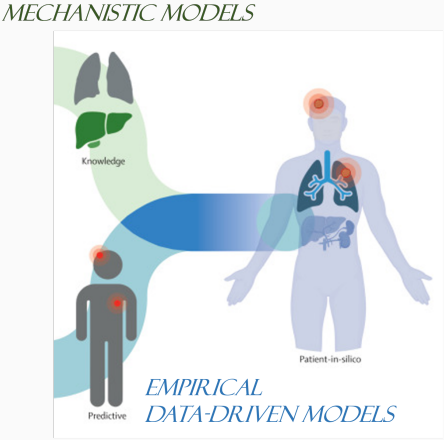


Mechanistic OR Empirical Models for Personalized Healthcare?



	Mechanistic	Statistical / ML
Basis	First principles	Data patterns
Data	Low (theory)	High (data-hungry)
Explain	High (causal)	Often low (black-box)
Example	Cardiac EP sim	AI retinal screen

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Key Takeaway

No single model type is superior. The best approach depends on the clinical question. Modern digital twins **combine multiple model types**.

The Tension: Population Models vs. the Individual Patient

Idiographic

idios (Greek: “one’s own”)

Particularizing

Understanding the **unique** individual

VS.

Nomothetic

nomos (Greek: “law”)

Generalizing

Discovering **universal** patterns

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Most of clinical medicine is nomothetic:
Guidelines, drug trials, risk scores—all derived from populations. They tell you what works *on average*.

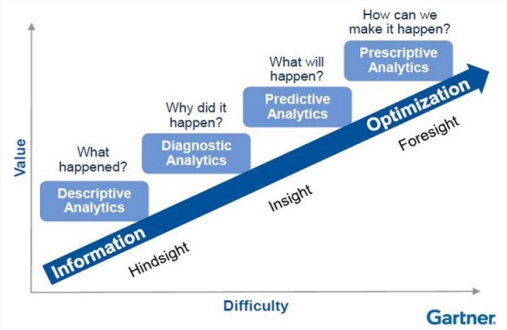
But your patient is not an average.
Unique genetics, comorbidities, social context, values.

Digital twins aspire to bridge this gap:
Start with population models (nomothetic), then personalize with individual data (idiographic).

The Promise

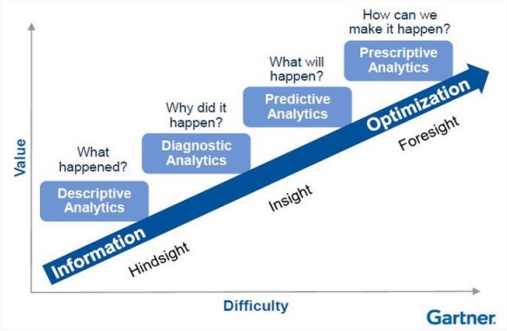
DTs could move medicine from “What works for most people?” to “What will work for *this* person?”

From Static Models to Living Models



Descriptive → Diagnostic → Predictive → Prescriptive analytics.

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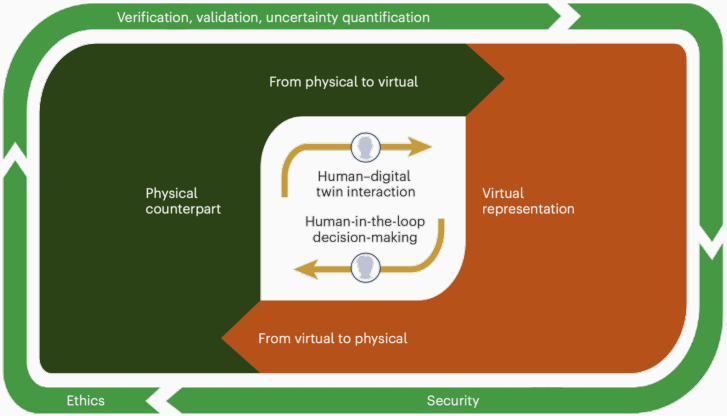
Descriptive → Diagnostic → Predictive → Prescriptive analytics.

- **Static model:** Built once, used to predict. *E.g., a risk score calculator.*
- **Simulation:** Run scenarios forward in time. *E.g., epidemic forecasting.*
- **Continuously updated model:** Learns from new data. *E.g., a learning health system.*
- **Digital Twin:** A *living* virtual counterpart dynamically linked via **bidirectional** data flow.

What is a Digital Twin?

A virtual counterpart dynamically linked to its physical system

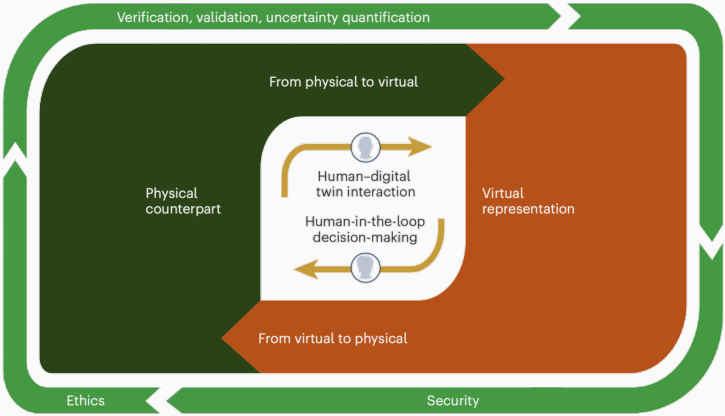
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  Continuously updates
- Predicts behavior
- Optimizes performance



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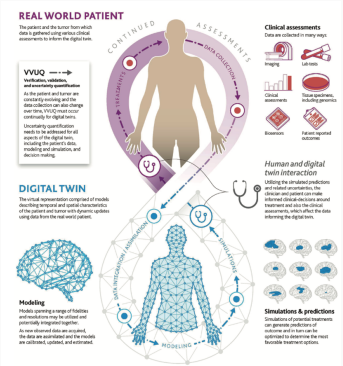
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A DT connects the physical and digital worlds through live feedback — representing, predicting, controlling, and prescribing — to transform data into understanding and purposeful action.

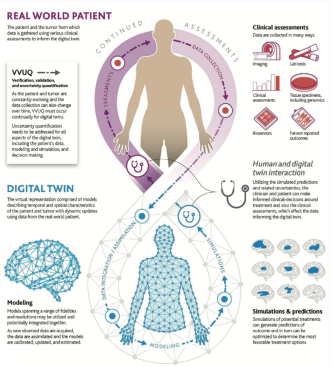
Digital Twins Across the Care Continuum



Example of a digital twin of a cancer patient and tumor

Credit: NASEM DT Report.

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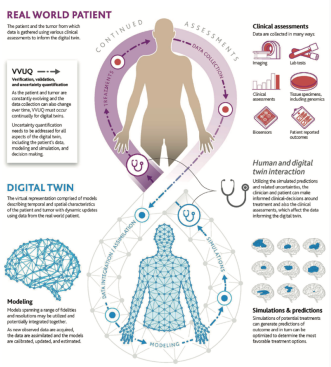
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Healthcare DT Applications

- **Personalized Diagnostics** — Physics-informed models for early detection
- **Treatment Optimization** — *In silico* pharmacokinetic trials for dosing precision
- **Hospital Operations** — Discrete-event simulation for patient flow and capacity
- **Precision Medicine** — Genomic–phenotypic fusion for targeted interventions

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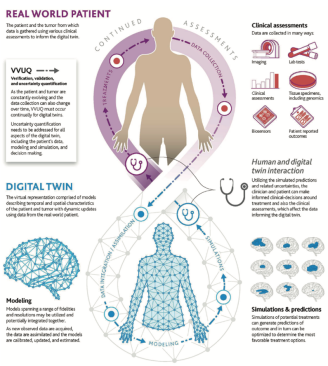
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Case Study: Oncology Digital Twin

Patient-specific tumor DTs integrate **imaging data**, **genomic profiles**, and **microenvironment dynamics** to simulate tumor progression and predict treatment response **before therapy**.

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From empirical dosing $\longrightarrow$ **Personalized precision oncotherapy**

How Modern AI Powers Digital Twins

Large Language Models (LLMs)
Tools like ChatGPT and Med-PaLM can synthesize unstructured clinical notes, summarize patient histories, and assist clinicians in interpreting DT outputs.

Foundation Models
Pre-trained on massive biomedical datasets, these models learn general representations of biology that can be fine-tuned for specific patients—accelerating DT personalization.

Computer Vision
Deep learning on radiology and pathology images feeds the DT's virtual representation with continuously updated anatomical data.

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Federated Learning
Train DT models across hospitals *without* sharing patient data—preserving privacy while improving predictions.

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Reality Check
AI models are powerful pattern finders—but they are not infallible. They hallucinate, they inherit biases from training data, and they do not *understand* causation. This matters enormously in medicine.

VVUQ: Can You Trust This Digital Twin?

Verification

"Are we building it right?"

Checking that the code correctly implements the mathematical model.

Validation

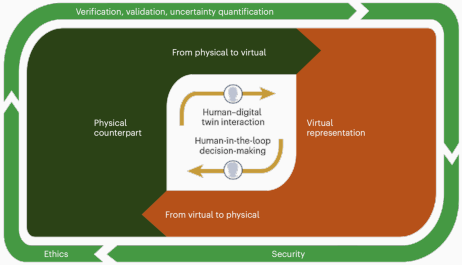
"Are we building the right thing?"

Comparing model predictions against actual clinical outcomes.

Uncertainty Quantification (UQ)

"How confident should we be?"

Quantifying uncertainty from incomplete data, model assumptions, and patient variability.



NASEM Recommendation

VVUQ must be **continually embedded** from the beginning, not bolted on at the end. Healthcare DTs require **dynamic validation** as patient conditions evolve and **rigorous VVUQ for AI/ML** components.

Activity: Design Your Own Patient Digital Twin (5 min)

Small Group Activity

In groups of 3–4, pick **one** clinical scenario:

1. A patient with newly diagnosed Type 2 Diabetes
2. A patient scheduled for cardiac surgery
3. An ICU patient with sepsis

For your scenario, sketch answers to:

- What **data** would feed the DT?
- What **models** would it use?
- What would it **predict**?
- How would clinicians **use** its output?
- What could go **wrong**?

Tips

Data sources: Wearables, EHR, labs, imaging, genomics, patient-reported outcomes.

Model types: Mechanistic (physiology), statistical (risk scores), ML (pattern detection), or hybrid.

Failure modes: Missing data, model bias, over-reliance, privacy violations.

We'll hear from 2–3 groups

Be ready to share your biggest insight *and* your biggest concern.

Ethical Dimensions of Patient Digital Twins

Privacy & Data Ownership
Continuous health monitoring generates intimate data. Who owns the “virtual you”? Who can access it?

Equity & Access
DTs require data, devices, and digital infrastructure. Will they widen the digital divide?

Consent & Autonomy
Can patients give meaningful informed consent for a continuously learning system? Can they opt out?

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Transparency & Explainability
If a DT recommends a treatment, the patient and clinician must understand *why*.

Accountability
When a DT-informed decision leads to harm: who is responsible?

> THERE ARE ETHICAL CHOICES IN EVERY SINGLE ALGORITHM WE BUILD



Privacy, equity, consent, autonomy, transparency, data ownership, inclusive design

Five Takeaways

❶ **Models are already central to medicine.**
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❝ Final Thought

"The map is never the territory. But a good map, held with humility, can help you navigate terrain you have never seen before."

Further Reading and Ongoing Work

Related Research on Healthcare Digital Twins

- Mekki, Y. M., et al. incl. **Qadir, J.** (2025). *Digital twins for the era of personalized surgery.* **npj Digital Medicine**, 8(1), 283.
- Alam, M., et al. incl. **Qadir, J.** (2025). *Digital Twins for Precision Oncology.* **Authorea.**
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